Forecasting Federal Reserve Policy Decisions Using Machine Learning

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***Abstract*-This project examines whether machine learning models can improve the forecasting of the Federal Reserve's monetary policy decisions. The task is formulated as a prediction of the policy interest rate at each FOMC meeting, using macroeconomic indicators available prior to the decision. The performance of several machine learning models is evaluated against an estimated Taylor rule, which serves as an economic benchmark.**

I Introduction

Monetary policy of the Federal Reserve (Fed) plays a central role in both the U.S. and global economies. Decisions made by the Federal Open Market Committee (FOMC) directly affect financial conditions, economic activity, and agents' expectations [1]. Although these decisions are infrequent, discrete, and closely monitored, they are not the outcome of a purely mechanical process. Setting the policy rate remains a key instrument of monetary policy, and forecasting it is both complex and economically crucial.

For financial markets and economic analysis, predicting the Federal Reserve's policy rate is of major importance. Changes in the policy rate directly affect borrowing costs, asset prices, and macroeconomic expectations. Anticipating monetary policy decisions is essential for investors and financial institutions in order to manage risk and price assets appropriately. From a forecasting perspective, predicting the policy rate also helps assess whether monetary policy follows a systematic response to macroeconomic conditions or varies across regimes, which is crucial for economists and policymakers.

This project is formulated as a regression problem aimed at predicting the level of the policy rate at each FOMC meeting. In order to avoid any information leakage, the analysis relies exclusively on macroeconomic information available prior to the decisions. Throughout the project, the performance of the Taylor rule is compared to that of several machine learning models, including penalized linear regressions (Ridge and Lasso) and nonlinear models (Random Forest and XGBoost).

The Taylor rule is used here as an economic reference to evaluate the performance of machine learning models. It proposes a simple relationship between inflation, economic activity, and the policy rate, and is widely used in the academic literature [3]. However, despite its analytical appeal, the Taylor rule relies on a rigid parametric structure and does not necessarily capture the full complexity of actual monetary policy decisions. It is therefore considered here as a benchmark rather than a predictive model sufficient on its own.

Given that Federal Reserve decisions may depend on nonlinear relationships, interactions between macroeconomic variables, and temporal dynamics, machine learning models appear to offer a flexible alternative that allows these patterns to be learned without imposing a strict functional form. However, the macroeconomic context poses important challenges for the application of such models, particularly related to the limited sample size, the risk of overfitting, and the need for rigorous out-of-sample evaluation.

II Research Question

II.1 General objective

The objective of this project is to assess whether machine learning models allow for better prediction of the level of the Federal Reserve's policy rate at FOMC meetings than an estimated Taylor rule.

II.2 Problem formulation

The research question is formulated as a regression task aimed at predicting the level of the policy rate set at each FOMC meeting. The objective is not merely to predict the direction of the change. This choice allows the analysis to capture both the direction of monetary policy decisions and their magnitude, making the exercise more informative from an economic perspective and more demanding empirically.

The Taylor rule is used as a point of comparison because it provides a standardized and widely adopted representation of monetary policy behavior [2][3]. It serves as a natural benchmark for evaluating whether more flexible models can improve predictive performance.

The objective is not to treat the Taylor rule as a complete description of FOMC decisions, but rather to use it as a reference to assess the potential improvements in predictive performance delivered by machine learning methods. These methods are able to adapt to more complex relationships between macroeconomic variables.

II.3 Motivation for using machine learning models

Monetary policy decisions may depend on nonlinear relationships, interactions between macroeconomic variables, and complex temporal dynamics [2]. Machine learning models allow these relationships to be learned without imposing a predefined functional form.

However, applying these methods in a macroeconomic context raises specific challenges, particularly related to the infrequency of decisions, the limited size of the available sample, and the risk of overfitting [4]. The central question is therefore whether the increased flexibility of machine learning models translates into a genuine improvement in out-of-sample predictive performance.

II.4 Scope of the analysis

The scope of this analysis is limited in several respects. First, the focus is placed on predicting the level of the policy rate at the time of FOMC meetings, rather than on long-term forecasting or policy scenario analysis. The objective is therefore not to model the entire monetary policy process, but to evaluate short-term predictive performance at discrete decision points.

Second, the analysis is restricted to information available prior to each FOMC meeting. This choice ensures a realistic forecasting framework and avoids any form of look-ahead bias. The study does not aim to incorporate real-time data revisions or forward-looking expectations beyond what is observable at the decision date.

Finally, given the relatively limited number of observations available at the frequency of FOMC meetings, the project prioritizes robustness and interpretability over model complexity. The comparison between the Taylor rule and machine learning models therefore places particular emphasis on out-of-sample performance and the risk of overfitting.

II.5 Contribution

This project aims to provide a rigorous empirical evaluation of the value of machine learning approaches for forecasting monetary policy decisions. By systematically comparing the predictive performance of linear and nonlinear models to a standard economic benchmark, it contributes to the debate on the relative relevance of complex models versus simple rules in a context of limited macroeconomic data.

III The methodology

III.1 Data and target variable

The first step in achieving the objective of this project was to collect data on historical policy rates corresponding to the different FOMC meetings. All data used in this project were obtained from the FRED database [5]. The data were divided into several subsets. First, a sample covering the period from 1982 to 2015 was used as the training set, meaning that it served as the basis for building the machine learning models. Second, a sample covering the period from 2016 to 2025 was used as the test set. The training sample was used to estimate policy rate values for the 2016–2025 period, while the test sample made it possible to evaluate whether these estimates were accurate and to compare the predicted values with the actual policy rate values observed at FOMC meetings and reported by FRED.

The variable to be predicted is the level of the policy rate set at each FOMC meeting. In order to cover a sufficiently long and consistent period, the DFEDTAR series (before 2008) and the DFEDTARU series (after 2008) are combined to form a homogeneous measure of the monetary policy rate. The change in the operational framework of monetary policy following the 2008 financial crisis represents an important methodological consideration. Prior to 2008, the Federal Reserve announced a single target rate, whereas since then it has communicated a target range for the federal funds rate. To ensure consistency of the target variable over the entire period studied, the policy rate is defined as the single target rate before 2008 (DFEDTAR) and as the upper bound of the target range after 2008 (DFEDTARU). Implementation details for the construction of the target policy rate series (DFEDTAR + DFEDTARU) are provided in *Appendix A*.

This choice is motivated by the fact that the upper bound of the corridor constitutes the main operational anchor of monetary policy [6]. The effective federal funds rate generally evolves close to this value. It is therefore conceptually equivalent to the single target rate used before 2008 and allows for the construction of a homogeneous monetary policy rate series over a long period.

The final sample includes approximately 180 decisions. This reflects both the low frequency of FOMC meetings and the usual sample size constraints encountered in macroeconomic applications [7].

III.2 Train-test split and evaluation design

The analysis adopts an out-of-sample prediction approach based on a strict temporal split [8]. The chronological train-test split is implemented as shown in *Appendix E*. Data covering the period from 1982 to 2015 are used as the training sample, while decisions taken between 2016 and 2025 constitute the test sample.

This choice makes it possible to evaluate the models' ability to generalize to an economic environment that may differ from the one observed during estimation. It also accounts for potential changes in monetary policy regimes, while respecting the temporal structure of the data. In order to avoid any form of overfitting, model performance is evaluated exclusively over the test period.

III.3 Feature construction and information timing

A central aspect of this methodology chapter concerns compliance with the information timing of monetary policy decisions. For each FOMC meeting, only macroeconomic information available prior to the decision date is used [9]. This constraint aims to replicate a realistic prediction framework and eliminate any look-ahead bias. The information-timing constraint is implemented via a backward merge-as-of join (*Appendix C*)

The explanatory variables are constructed from monthly series. Simple transformations are then applied, such as moving averages, variations, or lags. These transformations make it possible to capture relevant economic dynamics while maintaining an interpretable structure.

III.4 Model selection

The Taylor rule, estimated using linear regression, is used as an economic benchmark. It provides a simple and interpretable reference and is frequently used to analyze monetary policy behavior.

In addition, several machine learning models are considered in this project. These include penalized regressions such as Ridge and Lasso, as well as nonlinear models such as Random Forest and XGBoost. The choice of these models makes it possible to examine whether greater functional flexibility, despite the limited sample size, leads to an improvement in predictive performance.

IV Implementation details

This section describes the main steps of the project implementation, starting from the construction of the dataset to the training and evaluation of the models. The entire implementation was carried out in Python. In order to guarantee the reproducibility and readability of the code, I relied on standard libraries from the scientific ecosystem such as pandas, NumPy, and scikit-learn.

IV.1 Construction of the dataset

The first step of the implementation consisted in compiling a dataset that retraces the monetary policy decisions taken during FOMC meetings. Decision dates were identified based on changes in the policy rate, while the macroeconomic time series required for the analysis were extracted from the FRED database. Decision dates are detected as rate-change dates (*Appendix B*)

An essential aspect of the implementation concerns the temporal alignment of the data. Since macroeconomic variables are observed at a monthly frequency, while FOMC decisions are taken approximately eight times per year, it was therefore essential to ensure that only information available before each meeting was used. This alignment is achieved through a merging procedure, which associates each decision date with the most recent available observation of each macroeconomic variable. This approach makes it possible to avoid any form of look-ahead bias and guarantees a realistic prediction framework.

IV.2 feature engineering

Several simple transformations were applied to the raw macroeconomic series that are highly relevant from an economic perspective. These transformations aimed to improve the informational content of the features used by the models. They include three-month moving averages, designed to smooth short-term fluctuations, as well as first differences for certain variables. In addition, a lagged value of the policy rate was introduced in order to reflect the inertia generally observed in monetary policy decisions.

These feature engineering choices are deliberately parsimonious, given the limited size of the sample, while incorporating transformations that are widely used in the macroeconomic literature [10].

IV.3 Time frame and evaluation framework

The final dataset covers the period from 1982 to 2025 and includes approximately 180 FOMC decisions. In accordance with common practices in time series analysis, the dataset is strictly divided chronologically into a training sample and a test sample. Observations up to 2015 are used for model training, while the period from 2016 to 2025 is reserved for out-of-sample evaluation.

This choice makes it possible to assess the models' ability to generalize to more recent periods, characterized by different monetary policy regimes, while strictly avoiding any contamination between the training and test sets.

IV.4 Model training

Several models are trained and compared within a coherent empirical framework. The Taylor rule estimated by ordinary least squares serves as an economic benchmark. In parallel, several machine learning models are implemented, including penalized linear regressions (Ridge and Lasso) as well as nonlinear models (Random Forest and XGBoost).

For models that are sensitive to the scale of the variables, feature normalization is applied using a StandardScaler fitted exclusively on the training sample and then applied to the test sample. Scaling is performed by fitting the StandardScaler on training data only (*Appendix D*).This procedure guarantees that no information leakage occurs and allows for a fair and consistent comparison between models.

In the case of ensemble methods, conservative hyperparameter choices were adopted in order to limit the risk of overfitting given the limited size of the sample [4]. For the Random Forest model, hyperparameters were selected using a grid search combined with a cross-validation scheme adapted to time series data (TimeSeriesSplit). By contrast, the hyperparameters of the XGBoost model are fixed at reasonable values rather than being extensively tuned. This helps to avoid overfitting and to preserve out-of-sample robustness.

IV.5 Reproducibility

The entire pipeline is orchestrated from a single main script (main.py). It sequentially executes data construction, model training, performance evaluation, and the generation of result tables and figures. All outputs are automatically saved in dedicated files, which guarantees full traceability of the experiments.

This modular organization improves code readability and makes it possible to reproduce the complete set of results by running a single command [11].

V Code Maintenance

Maintaining clear, reliable, and extensible code is a central aspect of this project, given its empirical nature and the iterative development of the modeling pipeline. From the outset, the project was structured with an emphasis on modularity, version control, and reproducibility, following best practices commonly used in applied data science and empirical research.

V.1 Version control with Git

All source code is managed using Git, which allows for systematic tracking of changes throughout the development process [11]. Each major stage of the project was implemented incrementally and committed separately. These major stages include data construction, feature engineering, model training, evaluation, and visualization. This approach makes it possible to trace the evolution of the codebase, revert to previous versions if necessary, and clearly identify the contribution of each modification.

The use of a remote GitHub repository further facilitates transparency and reproducibility. It allows the entire project structure to be shared in a coherent and controlled manner, including scripts, configuration files, and documentation. Version control is therefore not only a technical tool, but also an integral component of the empirical workflow.

V.2 Modular code structure

The codebase is organized into distinct modules corresponding to the main stages of the analysis, namely data loading, feature construction, model estimation, and evaluation [12]. This modular design improves readability and simplifies maintenance, as individual components can be modified or extended without affecting the rest of the pipeline.

In particular, separating baseline models, ensemble models, and evaluation routines allows for straightforward experimentation with alternative specifications or additional models. This structure also facilitates debugging and reduces the risk of unintended side effects when updating the code.

For example, data preprocessing and feature construction are handled independently from model estimation, which makes it possible to modify the feature set without changing the training or evaluation routines.

V.3 Reproducibility and robustness

Reproducibility is ensured through a combination of modular design, fixed random seeds, and a clearly defined computational environment. In order to recreate the complete environment if necessary, all dependencies and library versions are specified in an environment configuration file.

In addition, the entire workflow is orchestrated from a single main script. This script sequentially executes data preparation, model training, performance evaluation, and result generation. This design makes it possible to reproduce the complete set of results by running a single command. It ensures consistency across experiments and minimizes the risk of human error.

This script guarantees that all results reported in the analysis are generated using a consistent data split, fixed random seeds, and identical preprocessing steps.

V.4 Testing and future extensions

While formal unit tests are not implemented given the scope of the project, robustness checks are integrated at key stages of the pipeline. These include explicit verification of data dimensions, checks for missing values, and validation of training and test splits. These safeguards help ensure the integrity of the results as the codebase evolves.

The current structure is designed to accommodate future extensions, such as additional models, alternative feature sets, or updated data. Thanks to the use of version control and modular design, these updates can be implemented in a controlled and transparent manner without compromising existing results.

Given the exploratory and empirical nature of the project, priority is given to transparency and robustness checks rather than to exhaustive unit testing.

VI Results

This section presents the empirical results obtained from the comparison between the Taylor rule benchmark and the machine learning models. The analysis focuses on out-of-sample predictive performance over the 2016–2025 period. This period was not used during model estimation. The results are presented using standard quantitative metrics as well as graphical comparisons between predicted and realized policy rates.

VI.1 Baseline models performance

I begin by examining the performance of the baseline linear models, namely the estimated Taylor rule, Ridge regression, and Lasso regression. These models provide a natural benchmark, as they rely on simple and interpretable relationships between macroeconomic variables and the policy rate.

From an economic perspective, the results suggest that a large share of the systematic variation in the Federal Reserve's policy rate can be captured by relatively simple linear relationships, once the variables are properly aligned in time and evaluated in a strictly out-of-sample framework. Penalization helps stabilize the estimates in a small-sample environment by limiting excessive sensitivity to noise.

VI.2 Numerical performance comparison

Model performance is evaluated using standard out-of-sample metrics computed over the 2016–2025 test period. The primary evaluation criterion is the root mean squared error (RMSE), as it penalizes large forecast errors. It is also particularly relevant in the context of monetary policy decisions, where large errors can have significant economic implications. For completeness, the mean absolute error (MAE) and the coefficient of determination (R^2) are also reported.

Model	RMSE (train)	MAE (train)	R^2 (train)	RMSE (test)	MAE (test)	R^2 (test)
TaylorRule OLS	0.372000	0.353000	0.946000	0.408000	0.288000	0.936000
RandomForest	0.240000	0.186000	0.978000	1.412000	1.100000	0.238000
Lasso	0.241000	0.179000	0.977000	1.433000	1.266000	0.216000
XGBoost	0.001000	0.001000	1.000000	1.473000	1.200000	0.171000
Ridge	0.238000	0.181000	0.978000	2.058000	1.928000	-0.619000

Table 1: Predictive performance of the models on the training and test samples

Table 1 highlights clear differences in predictive performance across models.

From a quantitative perspective, the Taylor rule clearly stands out as the best-performing out-of-sample model.

It achieves by far the lowest root mean squared error (RMSE) over the test period. It also attains a high coefficient of determination (R^2), indicating strong explanatory power and high predictive robustness.

Penalized linear models display more contrasted performance. Lasso achieves an intermediate performance, with an RMSE clearly higher than that of the benchmark but comparable to that of some more flexible models.

By contrast, Ridge exhibits a marked deterioration out of sample, with the highest RMSE and a negative coefficient of determination, suggesting an inability to generalize properly in this context. Nonlinear ensemble models, on the other hand, display a pronounced deterioration in out-of-sample performance.

VI.3 Ensemble models and overfitting

The analysis then proceeds with an examination of the performance of nonlinear ensemble models, namely Random Forest and XGBoost.

These models are designed to capture nonlinearities and complex interactions between macroeconomic variables. They therefore offer substantially greater functional flexibility than linear models. In the training sample, the performance of these models is extremely strong, particularly for XGBoost, which displays an almost perfect fit.

However, this strong in-sample performance does not translate into improved out-of-sample accuracy.

Over the test period, Random Forest and XGBoost exhibit RMSE values that are clearly higher than that of the benchmark, and their coefficients of determination remain low.

Although Random Forest achieves an intermediate performance among alternative models, it remains largely dominated by the Taylor rule in terms of predictive robustness. This substantial gap between in-sample and out-of-sample performance constitutes clear evidence of overfitting.

The limited frequency of FOMC meetings and the small sample size restrict the ability of highly flexible models to generalize reliably to new data, despite the use of conservative hyperparameters and time-series-adapted cross-validation schemes [4].

VI.4 Graphical comparison of predictions

To complement the numerical evaluation, Figures 1 and 2 present a graphical comparison between predicted and realized policy rates.

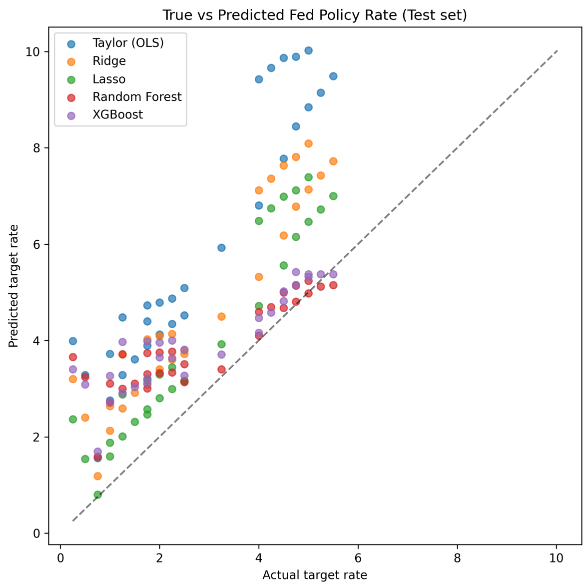

Figure 1: True versus predicted policy rates (out-of-sample)

At first glance, the true-versus-predicted scatter plot may suggest that several models, notably Lasso and some ensemble methods, produce visually better predictions than the benchmark over part of the sample. During relatively calm periods, these models appear to track the realized policy rate with comparable apparent accuracy.

However, this visual impression masks important differences that become evident during periods of rapid monetary policy changes.

When the policy rate changes abruptly—particularly during the post-2021 tightening cycle—more flexible models generate large deviations from realized values, especially at higher interest rate levels.

These deviations translate into a marked increase in out-of-sample errors.

By contrast, the Taylor rule, although it may appear less reactive in the short run, maintains a more stable predictive behavior during such regime changes.

Its economic structure limits extreme forecast errors, which explains its superior average performance as measured by RMSE.

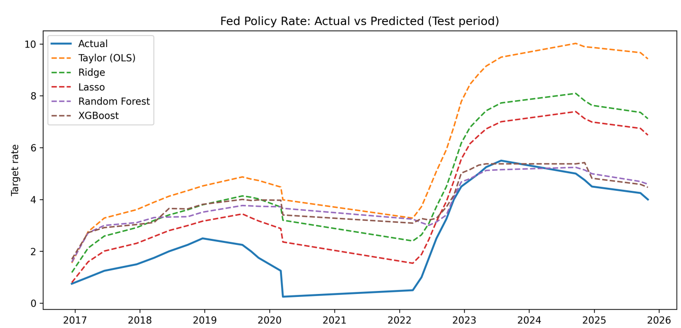

Figure 2: Time series of predicted and realized policy rates

The time-series comparison reinforces this conclusion.

Linear models closely follow the major trends in the policy rate over time. They retain relatively smooth dynamics even during easing and tightening cycles.

Ensemble models, by contrast, display more volatile behavior out of sample. They tend to overreact to local fluctuations, amplifying noise rather than signal.

VI.5 Main empirical takeaway

Taken together, the results indicate that simple and economically grounded models remain highly competitive for forecasting monetary policy decisions at FOMC meeting dates.

Although machine learning models offer increased flexibility in principle, this flexibility does not translate into systematic gains in out-of-sample performance in the studied framework.

In particular, more flexible models may appear to perform well during periods of gradual variation, but their accuracy deteriorates sharply during regime changes.

By contrast, the benchmark based on the Taylor rule exhibits superior robustness over the entire test period, leading to lower average forecast errors.

These results highlight the importance of model parsimony, rigorous temporal validation, and economic interpretability in low-frequency macroeconomic prediction applications.

VI.6 Discussion and limitations

The results reveal a clear trade-off between flexibility and robustness in monetary policy forecasting.

While some machine learning models produce visually convincing in-sample fits, their ability to generalize out of sample remains limited, particularly during episodes of high monetary volatility.

This pattern is especially visible during the post-2021 tightening cycle, where ensemble models generate large and persistent forecast errors.

Penalized linear models occupy an intermediate position, as they produce smoother predictions. Nevertheless, they fail to match the robustness of the economic benchmark.

Several limitations must be acknowledged. First, the limited number of FOMC meetings restricts the ability to reliably estimate highly complex models.

Second, the use of revised macroeconomic data may lead to a slight overestimation of predictive performance.

Finally, the analysis focuses exclusively on short-term prediction at decision dates and does not aim to anticipate longer-horizon monetary policy orientations.

VII Conclusion

This project evaluated whether machine learning models can improve the forecasting of Federal Reserve monetary policy decisions relative to a standard economic benchmark based on an estimated Taylor rule.

The results show that simple and economically motivated models systematically outperform more flexible models in terms of out-of-sample performance.

These findings highlight that, in the presence of low-frequency and limited macroeconomic data, robustness and interpretability may be more valuable than functional flexibility. They call for a cautious use of machine learning methods in monetary policy applications, with a strong emphasis on out-of-sample validation and economic coherence.

Appendix

A Construction of the policy rate target

To construct a consistent policy rate series over time, two FRED series are combined. The single target rate (DFEDTAR) is used prior to December 2008, while the upper bound of the target range

(DFEDTARU) is used thereafter. This choice follows the Federal Reserve's operational framework and ensures comparability over time.

```
cutoff = pd.Timestamp("2008-12-16")

df1 = df1[df1["date"] < cutoff]
df2 = df2[df2["date"] >= cutoff]

df_full = pd.concat([df1, df2], ignore_index=True)
df_full = df_full.drop_duplicates(subset=["date"]).sort_values("date").reset_index(drop=
    True)
```

B Identification of FOMC decision dates

```
df["prev"] = df["target_rate"].shift(1)
decisions = df[df["target_rate"] != df["prev"]].dropna().copy()
```

C Information timing and merge-as-of procedure

```
for name, series in macro.items():
        series = series.sort_values("date").reset_index(drop=True).copy()
        df = pd.merge_asof(
            df,
            series,
            on="date",
            direction="backward",   # take last value <= decision date
        )
```

D Feature scaling

```
    y_train = train["target_rate"]
    y_test = test["target_rate"]

    X_train = train.drop(columns=["date", "target_rate"])
    X_test = test.drop(columns=["date", "target_rate"])

    model.fit(X_train, y_train)

    y_pred_train = model.predict(X_train)
    y_pred_test = model.predict(X_test)

    ridge = Pipeline(
        [
            ("scaler", StandardScaler()),
            ("model", Ridge(alpha=1.0, random_state=0)),
        ]
```

```
    )

    lasso = Pipeline(
        [
            ("scaler", StandardScaler()),
            ("model", Lasso(alpha=0.05, random_state=0)),
        ]
    )
```

E Train-test split

```
train = df[df["date"] < "2016-01-01"].copy()
    test = df[df["date"] >= "2016-01-01"].copy()
```

F Use of AI tools

```
# AI Usage Disclosure

This document describes how AI-based tools were used during the development of this project, i

## Tools used
- ChatGPT (OpenAI)

## Purpose of AI usage
AI tools were used as support for:
- Clarification of methodological choices (e.g. fractionation of time series, prevention of da
- Improve the code structure
- Debug the Python code and check the implementation logic.
- Translation of the French-English report and help with the initial structure
- Help with English for oral preparation

## Scope and limits
- All the fundamental ideas, modelling choices, empirical design and interpretations are my ow
- AI tools were not used to generate the final code, results and conclusions.
- All the suggestions provided by AI were reviewed and adapted before their inclusion in the p

## Academic integrity
The use of AI tools complies with the course policy on AI-assisted work.
AI tools were used strictly as a support mechanism and not as a substitute for independent wor
```

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